

# Maturity and Yield Estimation of Tomatoes using RGB and Multispectral Images

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**Abstract**—Due to the rising cost of labor, smart agriculture is regarded as the trend of future agricultural development. In this work, we propose a system for maturity and yield estimation of tomatoes using RGB and multispectral images. The tomatoes are identified using object detection and tracking networks, followed by ripeness classification into the categories “mature”, “almost mature” and “immature”. In the experiment, the image sequence is used to detect tomatoes, and assign an ID to each of them. The principal component analysis is then performed using multispectral indices including NDVI, GRI and GNDVI. Finally, tomato tracking and maturity classification are integrated, and the number of tomatoes with various maturity in each group is counted. Through this research, we can more effectively monitor and manage tomato growth and improve agricultural production efficiency.

## I. INTRODUCTION

As the minimum wage continues to increase, the expense of employing seasonal labor for harvesting has steadily climbed. Moreover, this situation is exacerbated by the growing scarcity of available workers, which results in the competition of labor resources for farmers. When scheduling the workers to harvest, farmers need to maximize the cost-performance by considering the anticipated crop quantity or weight to be harvested. Thus, it is necessary to estimate the maturity and yields of the crop on a regular basis. To address this issue, image-based approaches are commonly used. Computer vision has played a crucial role in agriculture intelligence. Its applications extend beyond field monitoring, encompassing a range of techniques for inspection and classification of the crop. By employing the learning-based methods, several assessment factors such as size, color, shape, defects, and other attributes could be used to evaluate both the maturity and quality levels of crops.

Most of the existing methods for crop classification utilize the reflection of different wavelengths by fruits as a primary criterion. For instance, Guo *et al.* compared the wavelengths emitted by strawberries at various stages of maturity to classify their ripeness [1], while the blueberry ripeness was determined by assessing the wavelength of infrared light reflected from the fruit's skin [2]. The quality assessment can be conducted based on their changing appearances at different maturity stages, and use conventional classification techniques. In [3], blackberries and raspberries were categorized to 4 different maturity levels

(unripe, half-ripe, ripe, overripe) using CNN models including DenseNet, ResNet-18, and ResNet-50. Yang *et al.* [4] adopted Inception V3 to assess the maturity of individual strawberries at different stages, facilitating the classification and estimation of maturity progress throughout the strawberry's growth cycle with transfer learning.

In addition to quality and maturity assessment, vision-based techniques were also used to identify and locate ripe fruits for yield prediction and automated harvesting. For the localization of fruits, Liu *et al.* presented a multi-elliptical boundary model in Cr-Cb coordinate to separate the citrus fruits and tree trunks from background [5]. In a previous work [6], a robotic system capable of autonomously harvesting cherry was developed. It was equipped with thermal cameras to distinguish the infrared light reflection between ripe cherries and green leaves in order to identify the position of mature cherries more accurately. For yield prediction, Niedbala [7] employed a comprehensive array of factors as indicators of maturity. These included more than 30 multispectral indices, weather data spanning from the 121st to the 181st day of each year, and the foraging density of bees. Alternatively, DeLong *et al.* measured the chlorophyll in fruits weekly to predict the optimal time to start fruit harvesting [8]. By leveraging smart agricultural technologies, farmers are able to better manage orchards, take a timely action to prevent the occurrence and spread of pests and diseases, and enhance fruit quality and yield. Moreover, research on fruit yield prediction can provide valuable insights to the government in formulating better agricultural policies to promote agricultural production and development.

This paper focuses on the maturity and yield predictions of cherry tomatoes. Over entire harvesting seasons, three datasets consisting of both RGB and multispectral images are collected systematically. With a lateral camera movement, the tomatoes on the shelves of a tomato field are photographed for counting. For the identified tomatoes, their ripeness levels are classified based on RGB primary colors and multispectral indices. This study also analyzes the effectiveness of different classification indices and their impacts on the classification results. Finally, the quantities of tomatoes at each ripeness level in the image sequences are tabulated. It provides invaluable information for farmers in allocating labor resources during harvesting [9].

## II. PROPOSED APPROACH

The system flowchart of our proposed technique is depicted in Fig. 1. It contains the detection, counting, ripeness estimation, and ripening time prediction of tomatoes. The four key stages are then utilized to predict the yield of a tomato field. First, the RGB image sequence is fed into a detection network

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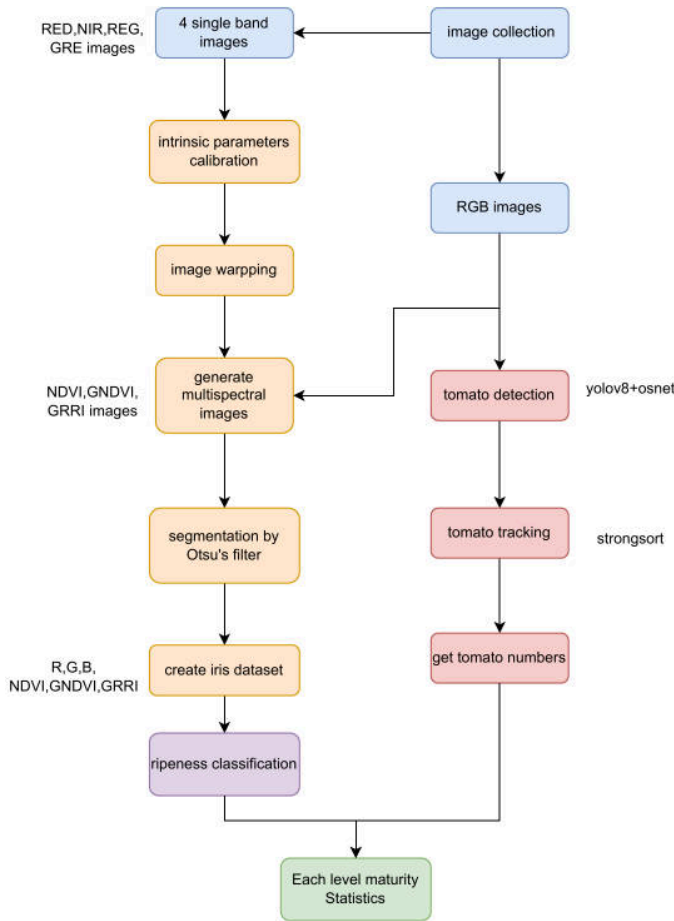


Fig. 1: The flowchart of our proposed system for maturity and yield estimation.

to identify tomatoes. Then, through a tracking network, each detected tomato is assigned a unique ID. With the image frame associated with each tomato, a ripeness classification method is employed to calculate the quantity of tomatoes in each ripeness stage. Finally, the ripening time of the individual tomatoes is estimated to achieve yield prediction of the orchard.

#### A. Classification Index

As crops progress through their growth stages, they reflect light of different wavelengths [10]. By capturing reflected light using multispectral cameras, various growth indicators of crop, such as leaf area, chlorophyll content, vegetation index, can be obtained. Based on the analysis of changes in these indicators, the maturity of crops can be determined. The indicators can be broadly classified using RGB images and multispectral images. In this paper, NDVI (Normalized Difference Vegetation Index) [11], GNDVI (Green NDVI), and GRRI (Green-Red Ratio Index) are utilized as indicators for maturity classification and predicting maturity time of tomatoes. Three algorithms, SVM (Support Vector Machine), KNN (K Nearest Neighbors), and ANN (Artificial Neural Network) are employed for classification of tomato maturity.

1) *NDVI*: The Normalized Difference Vegetation Index is an indicator used to measure land cover and vegetation growth. It is calculated based on the ratio of infrared and visible light reflectance obtained from remote sensing imagery [12]. The range of NDVI values is from  $-1$  to  $+1$ , where the negative values represent non-vegetated areas (such as rocks, bare land, water), while positive values indicate vegetated regions. In general, the higher NDVI value indicates more vigorous and healthier vegetation growth. NDVI is defined by

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (1)$$

where *NIR* and *RED* denote near-infrared and red channels, respectively.

2) *GNDVI*: The Green Normalized Difference Vegetation Index is another vegetation index adopted to assess vegetation health and density by analyzing the reflectance of green light within a given area. GNDVI values range from  $-1$  to  $+1$ , with higher values indicating healthier vegetation. It is defined by

$$GNDVI = \frac{NIR - GRE}{NIR + GRE} \quad (2)$$

where *GRE* denotes the green channel.

3) *GRRI*: The Green-Red Ratio Index defined by

$$GRRI = \frac{RED}{GRE} \quad (3)$$

aims to explore the feasibility of using airborne multispectral imagery to monitor the ripeness of coffee fields [13]. The spectral characteristics of the ground objects across different wavelength ranges can be captured by multispectral imagery. A series of airborne multispectral imagery data covering different growth stages and the degrees of ripeness in a coffee field are collected. Some image processing and analysis techniques are applied to convert this data into the GRRI index, which is used to assess the ripeness of coffee fields.

#### B. Classification Algorithm

After deriving the values of each tomato on various classification indicators, the next stage is to utilize SVM, KNN, and ANN algorithms to classify the ripeness level of each tomato. Under different parameter settings, SVM, KNN and ANN may be affected in terms of the error tolerance, generalization, and accuracy. In addition to the variation in parameter settings, the splitting of training and validation sets might also influence the classification results differently. For parameter settings, SVM's C value ranges from 10 to 40 with a step of 2 for classification, while the KNN's K value ranges from 2 to 30, also with a step size of 2 for classification. To ensure the classification stability, we adopt cross-validation in all algorithms. This will allow all training data to be trained and validated multiple times. Using cross-validation, all training data have a chance to be used as the testing set in rotation. For SVM, KNN, and ANN, we use 10-fold cross-validation for verification under each different parameter setting to obtain the most suitable parameters.

Finally, with the derived parameters, we classify the testing set to obtain the classification results. The detailed parameter

TABLE I: The parameter settings for each classification algorithm.

| Algorithm            | SVM                        | KNN     | ANN                      |
|----------------------|----------------------------|---------|--------------------------|
| kernel function      | rbf                        | -       | -                        |
| parameter            | C=10~40                    | K=2~30  | epoch=300<br>bat_size=40 |
| step                 | 2                          | 2       | -                        |
| cross-validation     | 10-fold                    | 10-fold | 10-fold                  |
| classification index | R, G, B, NDVI, GNDVI, GRRI |         |                          |

settings and the kernel functions used for the three classification algorithms are tabulated in Table I. The dimensions of the classification space depend on the number of indicators used for classification. For example, if only NDVI and GNDVI are used as classification indicators, the spatial dimension would be two.

### C. Tomato Detection, Tracking and Segmentation

To detect tomatoes in an image, we have utilized a detection network combining YOLOv8 and Osnet [14]. Osnet is a CNN model specifically designed for person re-identification, REID. It employs depth-wise separable convolutions, and replaces the original  $3 \times 3$  convolutional layer with a combination of a  $1 \times 1$  convolutional layer and a  $3 \times 3$  separable convolutional layer. In its Omni-Scale Residual Block, multiple bottleneck branches consisting of different sets of Lite  $3 \times 3$  convolutional layers are combined to form the multi-scale bottleneck. All bottlenecks are then aggregated using the Unified Aggregation Gate.

To track the individual tomatoes, we adopt StrongSORT and ByteTrack as separate tracking networks. The one with superior tracking performance is then selected as the tracking network for tomato quantity estimation in final analysis. StrongSORT [15] is improved over DeepSort [16]. The proposed ALink, an appearance-free linking model, converts short trajectories into complete trajectories. It takes spatio-temporal information of two trajectories as the input to predict their connectivity, and provides confidence scores for the connections using MLP. In ByteTrack [17], the low confidence issue caused by occlusion in multi-object tracking is addressed. An association method is proposed to retain and classify all bounding boxes into high-score and low-score. The low-score ones are then linked by Kalman filtering to predict the potential positions of targets in the next frame. This can effectively filter out the background, and resolve the problem of occlusion.

To remove the occluded branches, we refer to the work by Sari and Adinugroho for tomato segmentation [18]. Collected tomato images are first converted from the RGB to YUV color space. Otsu thresholding [19] is then applied to the V-channel for binary segmentation, removing the non-tomato regions and specular reflections. Due to dense tomato growth and possible occlusion by branches, leaves, and neighboring tomatoes, the rectangular bounding boxes are segmented into ellipses. This might include partially occluded or neighboring tomatoes, but it minimizes the potential error in pixel calculations caused by image segmentation.

It is required to compose NDVI, GNDVI and GRRI channels from multiple spectral images for analysis. However, the direct overlay without processing would result in misalignment and incorrect spectral values. To ensure the precise alignment of the same tomato in different spectral images, preprocessing is necessary for near-infrared, red, green, and red-edge images. In our implementation, it is a 4-step process given as follows. (1) the removal of lens distortion, (2) image warping to align the positions of near-infrared, red, and infrared images, (3) the alignment of images required for each multispectral index, and (4) application of the formulas for each multispectral index to the aligned images to obtain the corresponding TIFF images.

## III. EXPERIMENTS

### A. Data Collection and Ripeness Annotation

In the experiment we use a Parrot Sequoia+ camera for data collection. It is commonly used in agriculture applications, and composed of a multispectral sensor and a sunshine sensor. The multispectral sensor captures images in multiple spectral bands using five cameras on the sensor. In addition to RGB images, it captures NIR, red, green, and red-edge images, providing a more comprehensive vegetation image and data. The sunshine sensor acquires the state of sunlight illumination. Its purpose is to calibrate the multispectral images, and address variations caused by differences in sunlight intensity, to ensure consistent output images regardless of lighting conditions.

The captured tomato image sequences are transformed into the IRIS dataset format [20]. In the data format, the color data of each tomato in every frame, including the tomato ID, RGB images, multispectral images and ripeness, are recorded. The multispectral image data further includes NDVI, GNDVI, and GRRI. Except for the tomato ID and ripeness, the calculation for all other values involves averaging the pixel intensities of all unmasked parts within each tomato bounding box. In each dataset, all photographs of tomato image sequences, the IRIS format, the MOT dataset format<sup>1</sup> for the test image sequences, and the weights trained on the our data using SVM, KNN and ANN are included.

The grading of tomato ripeness is done on a weekly basis, by comparing data captured at the same location but different weeks to determine if the tomato is ripe and ready for harvest. If a tomato is not harvested and appears green in the  $n$ th week, remains unharvested and turns red in the  $(n+1)$ th week, and is harvested in the  $(n+2)$ th week, then it will be labeled as  $-2$  (i.e.,  $n - (n+2)$ ) in the  $n$ th week, and labeled as  $-1$  (i.e.,  $n+1 - (n+2)$ ) in the  $n+1$ th week. In addition, tomatoes that are more than three weeks away from ripening show minimal changes in color and shape. Thus, they will all be labeled as  $-3$  (including three weeks). In simple terms, we use  $-1$ ,  $-2$ ,  $-3$  to indicate the tomato is ripe and ready for harvest, taking one more week to harvest, and taking two or more weeks to harvest, respectively.

<sup>1</sup><https://motchallenge.net/instructions/>

TABLE II: The number of tomatoes in each dataset.

| dataset      | Training set | Testing set |
|--------------|--------------|-------------|
| 19th_tomato  | 4247         | 1775        |
| lin_tomato   | 4554         | 2423        |
| 19th2_tomato | 8720         | 4338        |

TABLE III: The classification results using different algorithms.

| Dataset      | Alg. | Classification Index |                  |                        |
|--------------|------|----------------------|------------------|------------------------|
|              |      | R,G,B                | GNDVI, NDVI, GRR | R,G,B,NDVI, GNDVI, GRR |
| 19th_tomato  | svm  | 63%                  | 70%              | 63%                    |
| 19th_tomato  | knn  | 64%                  | 66%              | 64%                    |
| 19th_tomato  | ann  | 62%                  | 73%              | 73%                    |
| lin_tomato   | svm  | 64%                  | 55%              | 65%                    |
| lin_tomato   | knn  | 58%                  | 58%              | 58%                    |
| lin_tomato   | ann  | 38%                  | 75%              | 46%                    |
| 19th2_tomato | svm  | 72%                  | 70%              | 78%                    |
| 19th2_tomato | knn  | 81%                  | 59%              | 81%                    |
| 19th2_tomato | ann  | 72%                  | 58%              | 73%                    |

### B. Ripeness Classification

By employing the approach for annotating tomato ripeness and utilizing the classification algorithms, the tomato ripeness in the test image sequences is classified. In the experiments, the classification accuracy is defined as the number of correct classification for tomato bounding box in all frames after the voting method divided by the total number of tomato bounding boxes in all frames. It is important to note that tomato tracking is not considered here, and all tomato bounding boxes in the training and testing sets are generated using the ground truth. The number of tomato bounding boxes in training and testing sets for three datasets is tabulated in Table II.

In image sequences, the same tomato can appear in multiple frames, potentially with the partial leaf obstruction or variation in the detected position. It may lead to inconsistent recognition results for the same tomato across frames. To address this, we employ the voting approach from ensemble learning. Multiple models are used to predict the target, and the most frequently predicted category is the final prediction result. The ripeness of the same tomato is determined by the majority of maturity classification. Table III tabulates the classification results. The accuracy is defined by the number of correctly classified over the total number of tomato bounding boxes in all frames.

In the results, it is observed that, except for the 19th\_tomato dataset, two other datasets showed better classification results in RGB compared to multispectral images. To figure out the superior performance of RGB classification, the contribution of each indicator is analyzed, as illustrated in Fig. 2. It shows that green light consistently provides the highest contribution to ripeness classification. The contributions of multispectral indices, red light and blue light varied among the datasets. It can therefore be inferred that better performance of the RGB classification is primarily due to the effectiveness of the green light component.

To further verify the correctness of inference, the difference

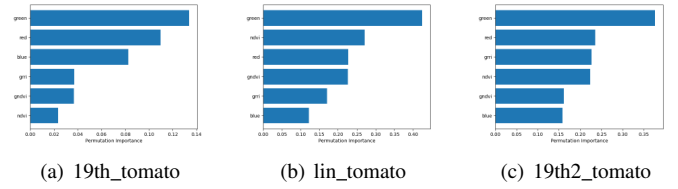


Fig. 2: The contribution of each indicator for the datasets.

TABLE IV: The classification results using only red and blue channels vs. RGB with all indices, where ‘N+G+G’ represents ‘NDVI+GNDVI+GRR’.

| Dataset      | Algorithm  | Classification Index |     |       |
|--------------|------------|----------------------|-----|-------|
|              |            | R+G+B                | R+B | N+G+G |
| 19th_tomato  | svm        | 63%                  | 61% | 70%   |
| 19th_tomato  | knn        | 64%                  | 62% | 66%   |
| 19th_tomato  | nn_network | 62%                  | 54% | 73%   |
| lin_tomato   | svm        | 64%                  | 44% | 55%   |
| lin_tomato   | knn        | 58%                  | 44% | 58%   |
| lin_tomato   | nn_network | 38%                  | 50% | 75%   |
| 19th2_tomato | svm        | 72%                  | 66% | 70%   |
| 19th2_tomato | knn        | 81%                  | 64% | 59%   |
| 19th2_tomato | nn_network | 72%                  | 66% | 58%   |

in classification performance is compared between RGB with all indices and only using red and blue channels. The results in Table IV shows that, except for the 19th\_tomato dataset, the classification performance of using RGB with all components is significantly better than utilizing only red and blue channels. Moreover, without the inclusion of green light as an indicator, the accuracy deteriorates even further compared to multispectral images. For the 19th\_tomato dataset, the influence of green light on classification is relatively smaller than the other two datasets. In Fig. 2, it can be seen that, although the contribution of green light still ranks first, the gap between green and the second-highest contributor is much smaller compared to the other two datasets. This could be why the 19th\_tomato dataset performs better in multispectral classification compared to the RGB.

### C. Change in Ripeness Annotation Cycle

In addition to comparing the classification performance, we also examine the accuracy and fuzziness of the classification algorithms on different maturity categories through the observation of the confusion matrix as shown in Fig. 3. It is found that the accuracy is better between -1 (mature) and -2 (almost mature), but much worse between -2 (almost mature) and -3 (immature). In the categories of almost mature and immature, their skin colors are very similar, making it difficult to classify based on the color.

Initially, the data collection cycle is once a week due to a considerable area of tomato fields to take care of. Through the observation during the collection of the first two datasets, it is found that until one week and a half before the tomato’s fully mature, there is almost no change in color. On the other hand, during the last one week and a half leading to maturity, there

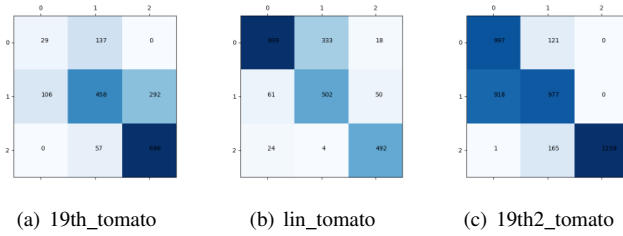


Fig. 3: The confusion matrix for classifying three datasets using all classification indices.



Fig. 4: The final output of detection and classification results.

is a drastic change in color. Thus, the third dataset is collected twice a week, instead of once a week.

#### D. Tomato Counting

The tomato counting process can be divided into two parts: detection and tracking. Initially, YOLOv8 is trained using the training set consisting of the RGB images captured by Parrot Sequoia's camera. The only class in the training set is tomato. Subsequently, the trained YOLOv8 model weight is passed to the tracking network. For the tracking network, the pre-trained models from StrongSort and ByteTrack are utilized. YOLOv8 is responsible for detection, while StrongSort and ByteTrack are responsible for assigning ID to the detection box. Finally, the allocated IDs are used to derive the number of tomatoes. The tracking results for three datasets are illustrated in Table V. It shows that StrongSort has the best performance in terms of MOTA and IDs, and is adopted for our yield estimation.

In the next experiment, we combine the tomato tracking and tomato maturity assessment. The objective is to calculate the number of tomatoes for each maturity level in the images, as well as their distributions. Table VI shows the statistics of the number of tomatoes with various maturity levels in all testing videos. Fig. 4 illustrates the final output of the detection and classification results. The tomatoes belonging to the "mature", "almost mature" and "immature" categories are represented by red, orange and yellow bounding boxes, respectively.

## IV. CONCLUSION

This work proposes a system to identify tomatoes and assess their ripeness levels using both RGB and multispectral images. Our approach consists of four parts: dataset establishment and ripeness labeling, tomato position tracking and quantity assessment, tomato ripeness classification, and statistical analysis of quantity of each ripeness category. We first employ YOLOv8 and OSNet object detection techniques to detect the tomatoes, followed by using StrongSort to assign appropriate IDs to all tomato bounding boxes. In the ripeness classification, principal component analysis is performed using various indices (NDVI, GRRI, and GNDVI). Subsequently, SVM, KNN, and Neural Network are applied separately to classify tomatoes into three classes: "immature", "almost mature", and "mature". Finally, the quantity and location of tomato in each ripeness category are obtained to assist farmers in planning harvesting routes and determining the optimal number of workers.

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TABLE V: The tomato tracking results.

| dataset           | algorithm  | MOTA   | MOTP  | FP  | FN   | IDs |
|-------------------|------------|--------|-------|-----|------|-----|
| 19th_tomato(6th)  | strongsort | 43.80% | 0.198 | 123 | 586  | 31  |
| 19th_tomato(6th)  | bytetrack  | 30.90% | 0.246 | 74  | 749  | 87  |
| 19th_tomato(7th)  | strongsort | 59.80% | 0.156 | 42  | 171  | 1   |
| 19th_tomato(7th)  | bytetrack  | 54.00% | 0.196 | 19  | 218  | 8   |
| lin_tomato(4th)   | strongsort | 7.60%  | 0.2   | 13  | 1394 | 3   |
| lin_tomato(4th)   | bytetrack  | 4.40%  | 0.22  | 0   | 1449 | 10  |
| lin_tomato(5th)   | strongsort | 21.40% | 0.177 | 39  | 497  | 1   |
| lin_tomato(5th)   | bytetrack  | 16.40% | 0.203 | 26  | 539  | 6   |
| lin_tomato(6th)   | strongsort | 23.30% | 0.139 | 59  | 405  | 3   |
| lin_tomato(6th)   | bytetrack  | 24.00% | 0.193 | 32  | 423  | 8   |
| lin_tomato(7th)   | strongsort | 9.80%  | 0.151 | 11  | 414  | 0   |
| lin_tomato(7th)   | bytetrack  | 8.70%  | 0.24  | 6   | 423  | 1   |
| lin_tomato(8th)   | strongsort | 30.70% | 0.175 | 49  | 197  | 0   |
| lin_tomato(8th)   | bytetrack  | 24.20% | 0.208 | 39  | 229  | 1   |
| 19th2_tomato(5th) | strongsort | 42.30% | 0.156 | 243 | 975  | 4   |
| 19th2_tomato(5th) | bytetrack  | 41.10% | 0.179 | 166 | 1081 | 1   |
| 19th2_tomato(6th) | strongsort | 40.70% | 0.163 | 193 | 1249 | 8   |
| 19th2_tomato(6th) | bytetrack  | 37.70% | 0.195 | 125 | 1383 | 15  |
| 19th2_tomato(7th) | strongsort | 47.90% | 0.163 | 193 | 1205 | 3   |
| 19th2_tomato(7th) | bytetrack  | 44.30% | 0.182 | 122 | 1362 | 12  |
| 19th2_tomato(8th) | strongsort | 49.20% | 0.147 | 156 | 1582 | 6   |
| 19th2_tomato(8th) | bytetrack  | 43.50% | 0.177 | 126 | 1792 | 19  |
| 19th2_tomato(9th) | strongsort | 75.80% | 0.137 | 86  | 66   | 2   |
| 19th2_tomato(9th) | bytetrack  | 76.10% | 0.189 | 69  | 80   | 3   |

TABLE VI: The final results of maturity and yield estimation.

| dataset           | ground_truth |    |    |       | prediction |    |    |       | Total number difference | RMSE |
|-------------------|--------------|----|----|-------|------------|----|----|-------|-------------------------|------|
|                   | -3           | -2 | -1 | total | -3         | -2 | -1 | total |                         |      |
| once a week       |              |    |    |       |            |    |    |       |                         |      |
| 19th2_tomato(5th) | 19           | 17 | 1  | 37    | 19         | 7  | 1  | 27    | -10                     | 5.77 |
| 19th2_tomato(7th) | 4            | 14 | 14 | 32    | 5          | 10 | 13 | 28    | -4                      | 2.45 |
| 19th2_tomato(9th) | 0            | 1  | 10 | 11    | 1          | 1  | 10 | 12    | 1                       | 0.58 |
| 19th_tomato(6th)  | 0            | 16 | 19 | 35    | 6          | 11 | 17 | 34    | -1                      | 4.65 |
| 19th_tomato(7th)  | 0            | 3  | 11 | 14    | 1          | 4  | 7  | 12    | -2                      | 2.45 |
| lin_tomato(4th)   | 16           | 11 | 2  | 29    | 1          | 6  | 0  | 7     | -22                     | 9.20 |
| lin_tomato(5th)   | 4            | 6  | 9  | 19    | 2          | 4  | 4  | 10    | -9                      | 3.32 |
| lin_tomato(6th)   | 4            | 4  | 12 | 20    | 1          | 4  | 5  | 10    | -10                     | 4.40 |
| lin_tomato(7th)   | 10           | 0  | 9  | 18    | 1          | 1  | 2  | 4     | -14                     | 6.61 |
| lin_tomato(8th)   | 2            | 4  | 1  | 7     | 3          | 2  | 1  | 6     | -1                      | 1.29 |
| twice a week      |              |    |    |       |            |    |    |       |                         |      |
| 19th2_tomato(5th) | 32           | 4  | 1  | 37    | 22         | 3  | 2  | 27    | -10                     | 5.83 |
| 19th2_tomato(6th) | 22           | 8  | 6  | 36    | 8          | 14 | 12 | 34    | -2                      | 9.45 |
| 19th2_tomato(7th) | 14           | 10 | 8  | 32    | 9          | 7  | 12 | 28    | -4                      | 4.08 |
| 19th2_tomato(8th) | 4            | 10 | 29 | 43    | 2          | 7  | 26 | 35    | -8                      | 2.71 |
| 19th2_tomato(9th) | 0            | 1  | 10 | 11    | 0          | 2  | 10 | 12    | 1                       | 0.58 |

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